AN OPEN IMMERSION OF LOCALLY RINGED SPACES FACTORS THROUGH THE ASSOCIATED OPEN SUBSPACE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 466-509, following the statement at 457-464.

Allowed standard prerequisites (editorial).

- Pullback of sheaves commutes with composition. Sheaves, Lemma sheaves-lemma-pullback-composition (cross-chapter, cited at ~474-475). Gives f^{-1}O_Y = f'^{-1} j^{-1} O_Y = f'^{-1} O_V.
- Isomorphism criterion for locally ringed spaces. lemma-isomorphism-locally-ringed (same chapter, cited at ~488). Turns the assembled pair (f', f'^sharp) into an isomorphism of locally ringed spaces.

Source reference. Original lemma 01HC: sources/stacks-schemes/pinned-original-source.tex, source lines 390–390. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 008I: focused-seeds/geometry/S9-schemes-open-immersion/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to Y$ be an open immersion of locally ringed spaces. Let $j : V = f(X) \to Y$ be the open subspace of Y associated to the image of f. There is a unique isomorphism $f' : X \cong V$ of locally ringed spaces such that $f = j \circ f'$.*

Proof. Let f' be the homeomorphism between X and V induced by f. Then $f = j \circ f'$ as maps of topological spaces. Since there is an isomorphism of sheaves $f^\sharp : f^{-1}(\mathcal{O}_Y) \to \mathcal{O}_X$, there is an isomorphism of rings $f^\sharp : \Gamma(U, f^{-1}(\mathcal{O}_Y)) \to \Gamma(U, \mathcal{O}_X)$ for each open subset $U \subset X$. Since $\mathcal{O}_V = j^{-1}\mathcal{O}_Y$ and $f^{-1} = f'^{-1}j^{-1}$ (Sheaves, Lemma 008I) we see that $f^{-1}\mathcal{O}_Y = f'^{-1}\mathcal{O}_V$, hence $\Gamma(U, f'^{-1}(\mathcal{O}_V)) \to \Gamma(U, f^{-1}(\mathcal{O}_Y))$ is an isomorphism for every $U \subset X$ open. By composing these we get an isomorphism of rings

$$\Gamma(U, f'^{-1}(\mathcal{O}_V)) \to \Gamma(U, \mathcal{O}_X)$$

for each open subset $U \subset X$, and therefore an isomorphism of sheaves $f^{-1}(\mathcal{O}_V) \to \mathcal{O}_X$. In other words, we have an isomorphism $f'^\sharp : f'^{-1}(\mathcal{O}_V) \to \mathcal{O}_X$ and therefore an isomorphism of locally ringed spaces $(f', f'^\sharp) : (X, \mathcal{O}_X) \to (V, \mathcal{O}_V)$ (use Lemma 01HC). Note that $f = j \circ f'$ as morphisms of locally ringed spaces by construction.

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Suppose we have another morphism $f'' : (X, \mathcal{O}_X) \to (V, \mathcal{O}_V)$ such that $f = j \circ f''$. At any point $x \in X$, we have $j(f'(x)) = j(f''(x))$ from which it follows that $f'(x) = f''(x)$ since j is the inclusion map; therefore f' and f'' are the same as morphisms of topological spaces. On structure sheaves, for each open subset $U \subset X$ we have a commutative diagram

$$\begin{array}{ccc} \Gamma(U, f^{-1}(\mathcal{O}_Y)) & \xrightarrow{\cong} & \Gamma(U, \mathcal{O}_X) \\ \downarrow \cong & \nearrow_{f'^\sharp} \quad \nearrow_{f''^\sharp} & \\ \Gamma(U, f'^{-1}(\mathcal{O}_V)) & & \end{array}$$

from which we see that $f'^\sharp$ and $f''^\sharp$ define the same morphism of sheaves. $\square$